Head Specialization: Run 2 Results

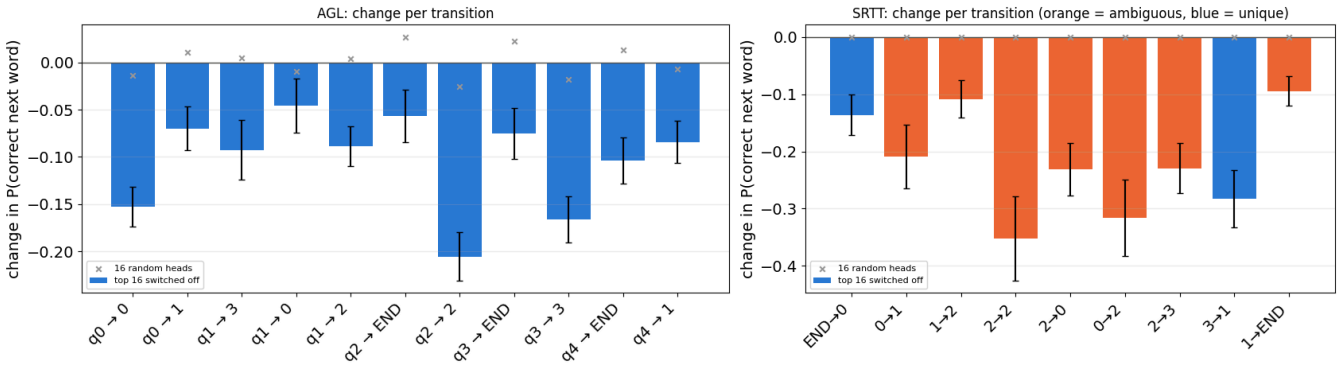
Nov 9, 2026 · @Dor

In SRTT the block-18 induction heads are interchangeable copies of one mechanism: any one of them restores most of the learning. In AGL they are partly divided, some pushing toward END and others toward continuing, and the learning needs their combination.

Part 1: switching off the top 16 heads, per transition

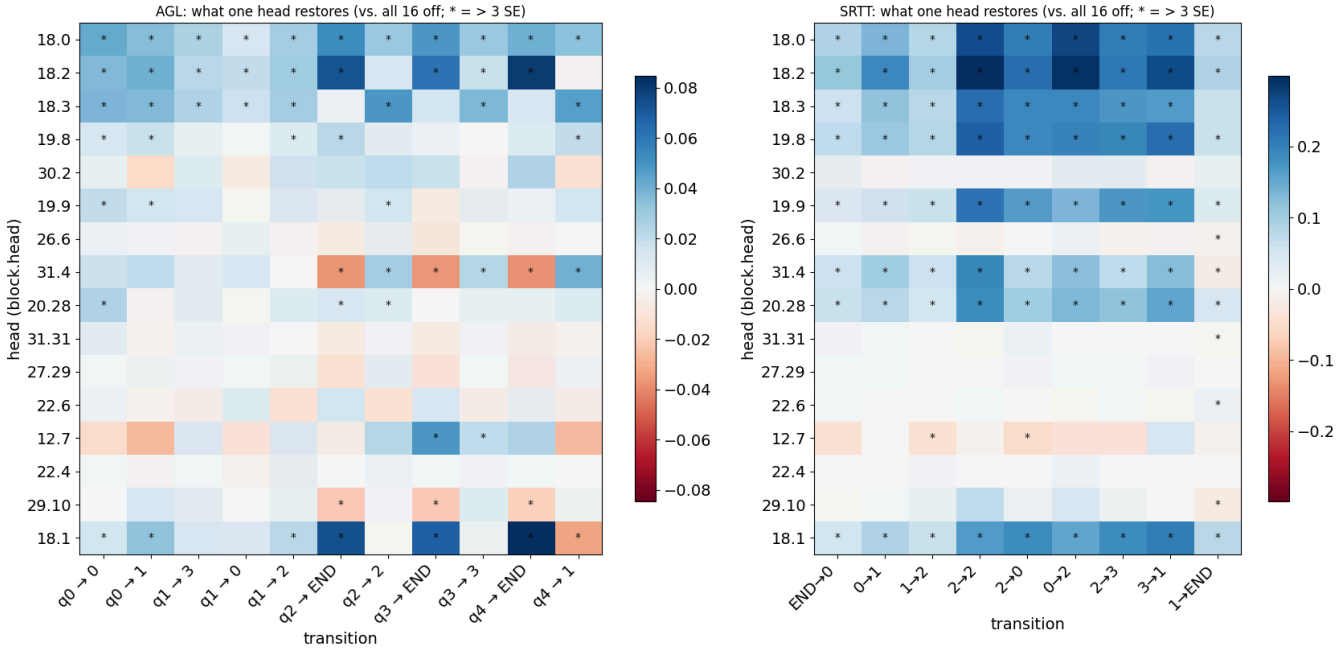
Every transition gets worse in both grammars, while 16 random heads from the same blocks change almost nothing. 20 paired runs per grammar.

- **SRTT: no ambiguity effect.** Ambiguous transitions drop 0.220, unique ones 0.210 (difference −0.011 ± 0.035). The hypothesis that these heads matter most where the previous word alone is not enough does not hold.
- **SRTT drops range from −0.09 to −0.35;** largest at 2→2 (−0.35), 0→2 (−0.32) and 3→1 (−0.28).
- **AGL: hardest hit are q2→2 (−0.21), q3→3 (−0.17) and q0→0 (−0.15).** The END transitions drop least (−0.06 to −0.10). Mean drop across the 11 transitions: 0.104.



Step C, SRTT: any single block-18 head restores almost everything

Leave-one-in: all 16 heads switched off except one; "restored" = learning with that head back on minus learning with all 16 off.



Head kept	Mean restored	Share of the 0.216 loss
18.2	0.198	~92%
18.0	0.172	~80%
19.8	0.153	~71%
18.3	0.141	~65%
18.1	0.131	~61%
19.9, 20.28, 31.4	0.088–0.117	40–54%
The other 8 heads	−0.02 to 0.02	~0

- **The restore is spread evenly across all 9 transitions** (specialization index ~0.07 for the block-18 heads). No head belongs to particular transitions.
- **One induction head is almost enough for a repeated sequence.** This explains the earlier result that removing the top 4 barely hurt SRTT (0.988): the remaining block-18 heads cover for them.

Step C, AGL: each head gives back little, and block 18 splits into END vs. word

The best single head restores only ~35% of the loss (18.2: 0.036 of ~0.104). AGL needs several heads working together, which fits the steady dose-response seen earlier (the top 4 alone already cost 21%).

Head kept	What it restores	Largest effect
18.1, 18.2	END transitions (q2/q3/q4→END)	+0.06 to +0.085; 18.1 slightly hurts q4→1
18.3	Word continuations (q2→2, q4→1), not END	+0.05
18.0	A little everywhere	+0.03 to +0.05
31.4	The opposite of 18.1/18.2: lowers END, raises words	−0.04 on END
Other heads	Little or nothing	—

The 18.2 = END / 18.3 = continue split from Step B (single-head ablation) is now confirmed with a second, independent test.

Caveats for presenting

1. **The AGL split is about the output type, not grammar states.** At q2 the choice is END vs. word 2, so boosting END automatically lowers the alternative. No head restores, say, q1 while leaving q3 alone. Say "heads differ in pushing toward ending vs. continuing", not "heads encode different rules".
2. **Ignore the specialization index for weak heads.** 12.7 scores 0.998 and 26.6 scores 0.716, but they restore ≈0, so the index is noise divided by almost nothing. Interpret it only for heads with a clear restore (block 18, 19.8, 19.9, 20.28, 31.4).
3. **12.7 alone does almost nothing** (slightly negative in SRTT, small for AGL q3→END). This fits the idea that block 12 feeds block 18 rather than acting alone, but this test cannot prove it.
4. **Leave-one-in measures what one head can do inside a damaged network.** The intact network may behave differently.

5. **Stars mark > 3 SE over 198 cells, with no multiple-comparison correction,** so a few isolated stars may be chance. The block-18 patterns are far beyond that.

Next test

Turn back on pairs (18.2 + 18.3), then all four block-18 heads, and check whether the AGL restores add up. That shows directly whether the END and continue heads are complementary.